MEKHI POWELL

mekhipowell25@gmail.com | linkedin.com/in/mekhipowell/ | Lakeland, FL | **Willing to Relocate** | Available May 2027

EDUCATION

Florida Polytechnic University **Expected May 2027**

B.S. Mechanical Engineering | Lakeland, FL

PROJECTS

Charpy Impact Tester – Design Lead **Feb 2026 - May 2026**

- Developed the complete Charpy impact tester assembly and dimensioned engineering drawings in SolidWorks, accounting for manufacturability, assembly, and a $100 project budget.
- Designed and 3D-printed a rotary encoder mount, iteratively refining its geometry to ensure proper fit, hammer clearance, and integration with the mechanical assembly.
- Integrated an Arduino Uno, rotary encoder, and LCD into the prototype; calibrated and tested the system to obtain repeatable impact energy measurements.

FPV Drone Frame – FEA Design Optimization **Jan 2026**

- Built a **parametric SolidWorks model** of an FPV drone frame and performed **FEA** under representative loading conditions to evaluate structural performance and identify high-strain regions.
- Refined frame geometry through iterative FEA, **reducing simulated maximum strain by approximately 78% and modeled mass by 6%** relative to the original design.

Aerodynamic Drag Testing & Instrumentation **Nov 2025**

- Designed and executed an aerodynamic test procedure to measure drag force and calculate drag coefficient across multiple airflow conditions using 3D-printed test bodies modeled in SolidWorks.
- Instrumented and calibrated a strain-gauged cantilever beam using LabVIEW and NI DAQ, then analyzed experimental results and documented measurement error sources and corrective actions.

FRC Robotics Team – Manufacturing Lead / Machinist **Sep 2020 – Jun 2023**

- Created dimensioned SolidWorks engineering drawings and machined robot components using **mills and lathes,** applying GD&T and verifying dimensional accuracy.
- Collaborated with CAD and assembly teams to resolve manufacturing constraints, prioritize critical components, and support robot assembly.
- Trained 10+ teammates in **mill and lathe operation**, bandsaw use, and sanding, emphasizing proper machining techniques and tool handling.

WORK EXPERIENCE

Engineering & Technology Instructor Intern **Jun 2026 – Jul 2026**

TechCORE2 + Urban League

- Demonstrated Onshape **CAD** techniques to 30 students, helping troubleshoot and refine **3D models** while reinforcing design intent, feature organization, and manufacturability.
- Proposed and introduced CAD instruction to the nonprofit, developing a zero-cost delivery plan, lesson structure, and instructional materials before leading 40+ classroom sessions.
- Collaborated with 5 other college interns to independently run daily program operations, coordinating 2 class sessions, dividing instructional responsibilities, and organizing engineering materials.

SKILLS

Manufacturing: Milling, Turning, Machining Fixture Use, Assembly, Tolerance Verification, 3D Printing, Material Selection

CAD & Drawings: SolidWorks, Onshape, Fusion 360, 3D Modeling, GD&T, Technical Drawings, Design for Manufacturing

Analysis & Testing: FEA, MATLAB, Femap, LabVIEW, NI DAQ, Calibration, Data Analysis

Programming & Electronics: Arduino/C, Soldering, Sensor Integration